1. You are designing a real-time feature extraction pipeline for pedestrian and vehicle detection in urban autonomous driving under night-time and rainy conditions.

Compare and critically evaluate:

• Edge detectors: Canny vs. Sobel vs. Scharr

• Interest-point detectors: FAST vs. Harris vs. Shi-Tomasi

• Descriptors: ORB vs. BRIEF vs. SIFT

Justify a combined pipeline (mention stages) by analysing trade-offs in speed, robustness to illumination/noise, and repeatability. Support your reasoning with mathematical principles (gradient magnitude/orientation, non-maximum suppression, binary tests) and discuss the role of multi-scale Gaussian pyramids.

2. Design an interactive segmentation pipeline for extracting a tumour from noisy 3-D MRI volumes that exhibit intensity inhomogeneity and weak boundaries. The pipeline must incorporate Active Contours (Snakes or Level-sets), Watershed, and Graph-Cut.

Explain the role of each technique, the order of application, and how user seeds/scribbles are used. Justify the design with respect to leakage prevention, topological correctness, and suitability for subsequent volume measurement.

3. Design a hybrid anomaly-detection scheme for high-dimensional IoT sensor streams. First discover clusters using DBSCAN or Hierarchical Clustering, then apply a supervised/semi-supervised classifier (One-Class SVM / Isolation Forest / Random Forest) together with an explicit decision boundary or reconstruction-error criterion.

Describe the sequential stages, the criterion for flagging a new sample as anomalous, and how the decision boundary is updated online. Justify choices regarding outlier robustness, scalability, and performance with very few labelled anomalies.

4. Design a complete dimensionality-reduction + classification pipeline for high-dimensional hyperspectral image patches that must support subsequent change-detection. Incorporate PCA or LDA or t-SNE together with a non-parametric classifier, then feed the output into a change-detection stage that uses Background Modelling and Optical Flow or Frame Differencing.

Describe stages, purpose of each technique, influence of the reduced space on background modelling, and how change parameters are estimated. Justify with respect to noise robustness, class separability, and near-real-time operation.

5. A remote-sensing application requires robust extraction of linear and textured structures (roads, river banks, crop patterns) from multi-resolution satellite images containing atmospheric noise and scale variations.

Design and justify a hybrid feature-extraction strategy integrating:

• Scale-space analysis (Image Pyramids + Laplacian of Gaussian)

• Gabor or steerable filters

• Discrete Wavelet Transform

• At least one of SIFT/SURF or HOG

Explain the contribution of each technique to multi-scale/multi-orientation representation. Critically compare the hybrid approach with using only Canny + Hough or pure Harris corners in terms of noise resilience, scale invariance, and discriminative power. Provide a high-level algorithmic outline and discuss limitations.

6. Compare the suitability of the following techniques for detecting circular and elliptical objects (traffic signs, manhole covers) in real-time traffic-camera video under varying illumination:

• Circular Hough Transform vs. Randomised Hough Transform

• Ellipse fitting (Direct Least Squares vs. RANSAC)

• Contour-based methods (Suzuki–Abe vs. Douglas–Peucker)

Propose a combined pipeline, justify stages with mathematical principles, and analyse trade-offs in computational cost, robustness to partial occlusion, and localisation accuracy. Discuss how multi-resolution processing can improve performance.

7. Design an interactive 2-D/3-D segmentation pipeline for extracting blood vessels from noisy angiographic CT volumes with thin structures and intensity variations. The pipeline must use Region Growing, Hessian-based vesselness filtering, and Graph-Cut or Random Walker.

Explain the role and sequence of each technique and the use of user-provided seeds. Justify the design regarding leakage control, preservation of thin structures, and topological correctness for subsequent centreline extraction.

8. Design a hybrid pattern-recognition scheme for fault detection in high-dimensional industrial vibration sensor data. First perform unsupervised grouping with Gaussian Mixture Models or K-Means++, then employ a supervised classifier (Bayes / SVM / MLP) together with an explicit Mahalanobis-distance or residual-based discriminant function.

Describe sequential stages, the decision rule for declaring a new sample anomalous, and how the discriminant is constructed/updated. Justify with respect to outlier robustness, computational scalability, and operation with scarce labelled fault examples.

9. Design a dimensionality-reduction and tracking pipeline for high-dimensional feature vectors extracted from multi-camera surveillance video. Incorporate ICA or PCA or Autoencoder-based reduction together with a non-parametric density estimator, then feed the result into a motion-analysis stage using Background Subtraction and Lucas–Kanade or Farneback Optical Flow.

Describe stages, purpose of each method, effect of the reduced space on background modelling, and estimation of motion parameters. Justify regarding noise robustness, preservation of discriminative information, and real-time suitability.

10. A dermatological imaging system must extract textured skin lesions and oriented structures (hair, vessel patterns) from multi-scale dermoscopic images that contain specular noise and scale variations.

Design and justify a hybrid feature-extraction strategy that integrates Scale-Space (Gaussian pyramids + derivatives), Gabor filters, Discrete Wavelet Transform, and HOG or LBP.

Explain the contribution of each component. Critically analyse advantages over single methods (Canny, Harris, pure Hough) in noise resilience, scale/rotation invariance, and discriminative power. Provide an algorithmic outline and discuss limitations.

11. You are designing a feature pipeline for lane-marking and road-sign detection in adverse weather (fog, rain, low light). Critically evaluate:

• Edge detectors: Canny vs. LoG vs. Phase Congruency

• Line detectors: Progressive Probabilistic Hough vs. LSD

• Corner/interest-point detectors: Harris vs. Hessian-Affine vs. AGAST

• Descriptors: SIFT vs. SURF vs. HOG

Propose a combined multi-stage pipeline, justify trade-offs (robustness to illumination/scale, speed, localisation), and support reasoning with scale-space theory, gradient histograms, and multi-scale filtering. Discuss further gains from Image Pyramids or steerable filters.

12. Design a complete interactive segmentation pipeline for extracting an organ with fuzzy boundaries from noisy ultrasound volumes. Incorporate Marker-controlled Watershed, Active Contours, and Graph-Cut.

Explain roles, sequence, and utilisation of user scribbles/seeds. Justify the design for preventing over-segmentation, controlling leakage, and producing topologically correct masks suitable for subsequent shape analysis.

13. Design a hybrid anomaly-detection framework for high-dimensional network-traffic feature vectors. First discover natural groupings with K-Medoids or Spectral Clustering, then apply a semi-supervised classifier (Label Propagation / One-Class SVM / Autoencoder) together with an explicit reconstruction-error or density-based discriminant.

Describe stages, the criterion for anomaly declaration, and online updating of the discriminant. Justify choices concerning outlier robustness, scalability to streaming data, and performance with extremely few confirmed attack examples.

14. Design a dimensionality-reduction + motion-analysis pipeline for high-dimensional multi-spectral image sequences that must support object tracking. Use LDA or PCA or NMF together with a non-parametric method, then apply Background Subtraction and dense Optical Flow.

Describe sequential stages, purpose of each technique, influence of the reduced feature space on background modelling, and estimation of motion parameters from foreground regions. Justify with respect to noise robustness, class separability, and real-time constraints.

15. A materials-science application requires robust extraction of oriented microstructures and textured regions from multi-resolution SEM/TEM images containing significant noise and scale variations.

Design and justify a hybrid feature-extraction strategy integrating:

• Scale-space analysis (pyramids + Gaussian derivative filters)

• Gabor or Log-Gabor filters

• Discrete Wavelet or Curvelet Transform

• At least one of SIFT/SURF or HOG

Explain how each technique contributes to multi-scale and multi-orientation representation. Critically compare the hybrid approach with using only Canny, Harris, or pure Hough Transform regarding noise resilience, scale invariance, and discriminative power. Include a high-level algorithmic outline and discuss potential limitations.